Healing Group

وعزمنا أن نحسن سعيا

Micro Lab Mid

2017

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كلية التمريض

Micro Lab

*** All work in micro lab must be under aseptic condition:**

1- wear your lab coat " closed".. Why ? :

- To prevent infection of our self .
- To prevent any accidental discoloration of our clothing by staining .

2- Close all source of air .. Why ?

- To prevent pollution of outdoor air .
- To prevent contamination of our culture .

3- wash your hand using soap and tap water to wash dust then sterilize them using Antiseptic like --> Hygiene (which use for living tissues) .

4- Disinfect the bench using a disinfectant like (70% OH) → (which use for non-living tissue)

- To prevent infection of our cell
- To prevent contamination of culture .

5- sterilize the air surrounding the working area by heat

(we sterilize 1 feet around benzene burners)

* Tools and materials used in micro lab :

1- Slide -->

- Wash the slide with soap and tap water (**to remove dust and grease**)
- **Rinse** the slide with disinfectant (to kill **all** type of microorganism)
- Dry the slide near **heat** or by **filter paper** .

2- Sterile – distilled water (St.d.H2o) → No Salt , No traces , ionize (have all ions except CL) → because it kill bacteria .

* <u>sterile</u> → By out clear (120 c , 15 min , 1 micro par pressure)

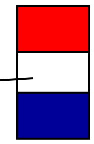
*** H2o + AgNo3 → AgCl2 (white)**

- (not sterile water) → **Cl لانه يوجد**

3- microbiology loop (——o) " sterilize it by heat in colorless area "

4- Microbiology needle (——) " sterilize it by heat in colorless area "

5- Benzene burner → have 3 zones " Red , colorless , Blue "



أقوى منطقة دائما التعقيم فيها

6- Cotton swap .

7- Microscope → Light Microscope

*** Decrease in distance , increase in magnification .**

1- Eye lenses →(10x) .

2- Objective lenses (4x , 10x , 40x , 100x)

4x → Low power .

10x → medium power .

40x → High power " can see bacteria " .

100x → Oil immersion lens .

3- Coarse adjustment (focus) → 40x

4- Fine adjustment (perfect focus) → 100x only

Microscope

يجب قراءة اول لاب بالمانيوال
لرؤية صوره ومعرفة اجزاءه

****

*** Wet mount slide :**

used to study living cell (bacteria) …Why living ?

- To study behavior , movement of bacteria (motility)

- To study natural size and shape of bacteria .

*** Procedure :**

1. Prepare clean, sterile, and completely dry slides .

2. By sterile loop add (3) loopful of (St.d.H2o) to slides .

3. Resterilize to loop, and cooling in empty area in culture.

(prevent scattering of bacteria in door – prevent infection of our cell)

- Because high temp → kill bacteria .

4. Put it in slide (small amount) and mix with water until milky suspension

5. Add cover slip (water بنزل بزاوية والطرف على اخر طرف)

6. Exam your slide under microscope .

* **Bacteria is colorless** but the media consist of **(glucose, fructose**, **O2, amino acid)** has a **milky color** → because he have a big amount of it .

Bacteria

- **Motile** → **Movement in all direction**
- **Non-motile** →
  - **- Fixed**
  - **- Movement in one direction**
  - **- Brownian movement " حركة اهتزازية "**

*** (All bacteria is colorless, transparent and minute)**

*** Bacteria staining ***

Stain → **Benzene** ring **+ chromophone** →(compound impart color to benzene)

Benzene ring → Organic compound (color less)

Give ↓

Auxochrome + chromogen (Not stain) → **when added auxochrome to chromogen the result is Dye**

- Auxochrome → **Impart charge to chromogen**
- chromogen → **Color organic compound**

Dye (according to charge)

- **(+) Basic dye**
- **(-) Acid dye**

* Acid dye like → Nigrosine (black)
* Basic dye like → Crystal violet , safranin (pink) , methylene blue (blue) , malachite green (green) , carbol fuchsine (red)

* Net charge of bacteria is Negative → on cell wall

* Basic dye is perfect to use

Bacteria + Basic dye ⊕ → attraction (can penetrate cell)

Bacteria + Acid dye ⊖ → repulsion (can't penetrate) (Just stain the back ground)

* The dye is for the cytoplasm

*According to our purpose there are 3 stain techniques:

1- simple stain (one dye that reacts similarly to all bacteria) .

Direct → Stain the bacteria (Basic dye)

Indirect (Negative stain) → Stain the background (Acid dye)

2- Differential stain => (use two or more dye)

That react **differently** with different dye .

- **Gram stain**
  - +
  - –
- **Acid-fast stain**
  - Acid fast
  - Non-Acid fast

3- Structural stain : we use it to determine the specific structures of bacteria

(capsule staining, spore staining)

Simple stain

(use for shape ,size ,arrangement)

1. Direct stain => (Basic dye) → For shape and arrangement.

1-smear preparation

(in all types of dying have smear preparation except indirect stain)

*step of smear preparation:

- clean, sterile, complete dry slide .

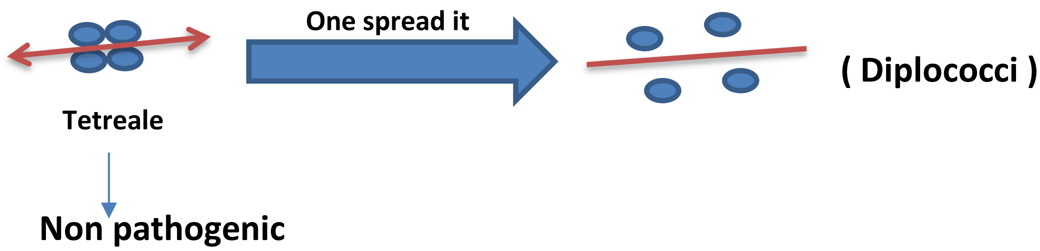
- Add 3 loopful of st.d.H2o

- Resterilize the loop , check it cooling , and put bacteria and mix it until milky suspension .

* Spread the mixture over proper area on slide

- نحدد المساحة حسب كمية البكتيريا وهي خطوة مهمة " لو لم نفعلها سيتغير ترتيب البكتيريا "

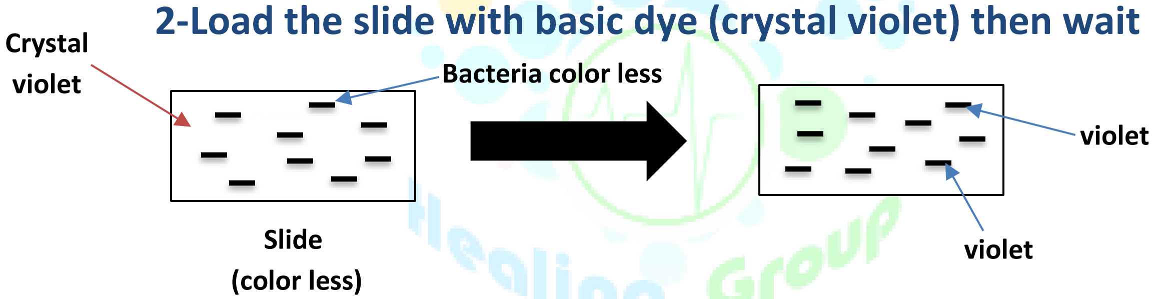
(70% of bacteria arrangement should appear as in nature)



- Air drying near benzene burner
- Heat fixation by pass it six time over the benzene burner

(No to kill bacteria . it just to avoid remove during stain)

2-Load the slide with basic dye (crystal violet) then wait 1 min



3-Wash the slide under tap water (to remove excessive amount of stain) But avoid direct harsh washing

4-Blot the slide gently by filter paper

we should see colorless background and violet color to bacteria.

2. Indirect stain (negative)

1-To determine size

2- To stain bacteria can't be stained by direct

acid dye (negative) → it very toxic

we use large molecular weight compound (ink)

*No smear preparation

* procedure :

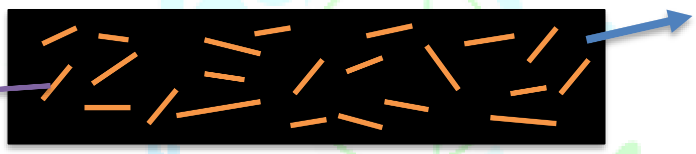
1-prepare 2 clean, sterile, dry slides .

2- Add small drop of ink to slides at edge

3- By sterile cool loop , add loopful of bacteria , then mix it with ink .

4- spread the mixture in <u>one direction</u> (unidirectional) use other slide

5- Air drying



------------------------| Differential staining |--------------------------

* gram stain :

- According to cell composition :

1- Gram negative (G-ve) → mainly composed of lipopolysaccharide (lps)

2- Gram positive (G+ve) → peptidoglycan

3- Acid fast → mycolic acid (wax)

* There are <u>3</u> methods to distinguish between G+ve and G-ve :

1. Selective media :

media contain selective chemical allow some bacteria to grow and limit other type :

-EMB → Eosin – methylene blue
- MacConkey
- TSI → triple sugar iron

→ Selective for (G – ve)

-EMB
-TSI
-MacConkey

\+ Bacteria

Grow → (G-ve)

Don't grow → (G+ve)

Incubation 37°C (for 24 hours)

* But this method isn't 100% efficient because the bacteria has high mutation rate to adapt

2. (10% KOH) : can dissolve LPS but not peptidoglycan or mycolic acid

This method isn't 100% efficient because can't differentiate between gram (+) and acid fast

1- Add small drop of 10% (KOH)
2- By sterile , clean , cool add loopful of bacteria and mix it

(Not mucoid)

No LPS
(could be gram (+ve) or acid fast).

1- Add small drop of 10% (KOH)
2- By sterile , clean , cool add large loopful of bacteria and mix it

(Mucoid)
(white thread)
(- ve) بتكون

* E.coli → - ve small size

* Bacillus → + ve large size

3. Gram stain :

* هنا يتم تحضير الـ٣ شرائح الاولى والثانية مثل ما تعلمنا بتحضير الشرائح سابقاً
* رح نشرح الشريحة الثالثة فيها اضافات لانها دمج للشريحتين

1- smear preparation :

1. E.coli 2. Bacillus 3. Mixed

A- clean, sterile , dry slide

B- By sterile loop , add 3 loopful of **St.d.H2o** to the center of slide

C- resterilize loop and check it cooling (in **E.coli plate**) then add a **large** loopful of E.coli and mix it until milky suspension .

D- resterilize loop and check it cooling (in **bacillus plate**) then add **small** loopful of bacillus then mix it very well

E- spread over large area on the slide

F- Air drying

G- Heat fixation

2- Load the slides with primary stain (crystal violet) for 1 Min

3- Wash the slide under tap water (to remove excessive)

4- Load the slides with mordant (مثبت) (Iodine) for 1 Min

E.coli

Crystal RNA

Iodine
↓
Bind with crystal
↓
complex

All gram negatives not have mg^{2+}
↓
Crystal - I2 - complex

Bacillus

Crystal RNA

Iodine
↓
Bind with crystal
↓
complex

All gram positive has mg^{2+}
↓
Crystal - I2 - Mg - RNA - complex

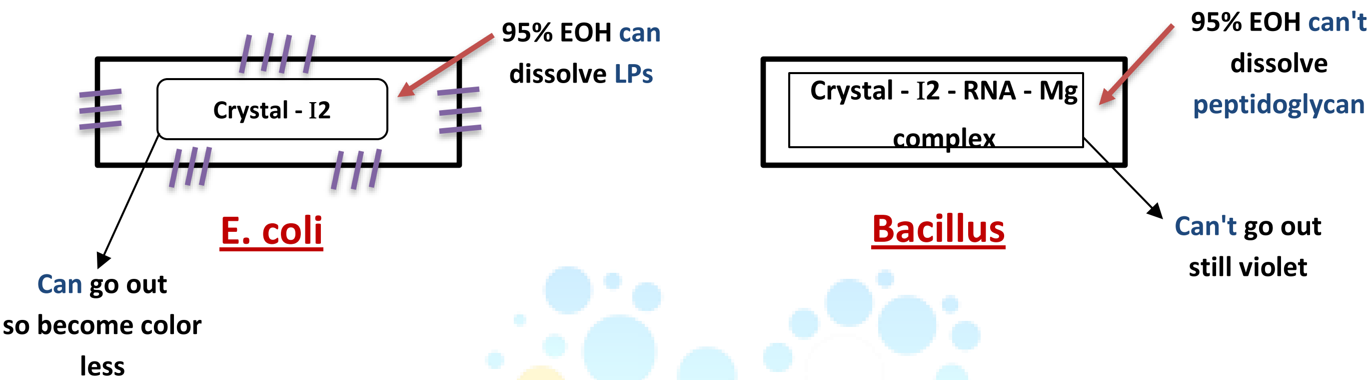
So Iodine fix the primary stain inside both of cells

5- Wash slide under tap water (to remove excessive amount)

6- Wash the slide by decolorization solution (95% ethanol or absolute)

Can dissolve LPS but not peptidoglycan or mycolic acid

7- Wash the slide immediately under water (to stop action of alcohol)



* If we don't wash the slide after 95% EOH , Bacillus will be affected because the cell wall will break down and it became colorless .

This is called (over decolorizing)

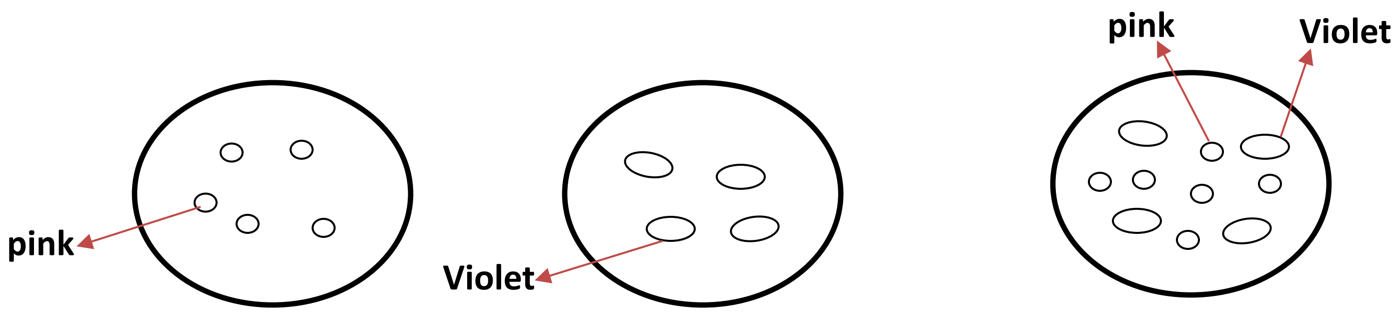
* While if we use little of 95% EOH , E. coli will be affected because it remain violet .

This is called (down decolorizing)

8- Load the slide by counter stain (safranin) for 1 Min .

9- Wash the slide under tap water (to remove excessive)

10- Blot the slide by filter paper .



E. coli (small size) | Bacillus (large size) | Mix

* we can't change between primary stain and counter stain
* we can change between crystal violet only with methylene Blue

* If we get pink G+ve bacteria then we had mistaken and over decolorizing
* If we get violet G-ve bacteria then we had mistaken and down decolorizing

* Gram stain culture to be used must be fresh (less than 24 hours)

* Old culture → cell wall become more fragile and can break

------------------------------| Acid - fast |---------------------------------
(color less)

* Consider as (gram + ve)

gram + ve

Acid – fast
(color less in gram stain)
(mycobacterium)
نفس شكل
E. coli
Small, single arrangement
short bacilli shape

Non - Acid fast
(bacillus)

***All acid- fast is gram positive but not all gram positive is acid-fast**

* Acid – fast stain :

1- (smear) preparation :

Note

هنا نفس خطوات اعداد ال (smear) في التجارب السابقة
مع مراعاة اختلاف انواع البكتيريا

mixed → mycobacterium (large amount **) + Bacillus (**small amount **)**

- Fresh culture less than **24** hours .

2- Load the slide with primary stain (carbol fuchsine) for 5 Min under staining (avoid boiling and dry)

Carbol fuchsine

Bacillus

Heat

Carbol fuchsine
Can't penetrate because of wax so we add heat (steam)
1.dissolve wax
2. Increase solubility to carbol fuchsine in mycobacterium

mycobacterium

Heat

3- cool the slide at room temp for 5 Min to mordant the stain

- harden the wax
- decrease the solubility

4- Wash the slide with decolorizing solution (acid – alcohol)

(3% HCl + 95% EOH)

Very strong decolorize
dissolve carbol fuchsine

Acid-alcohol

Red color

Bacillus

Color change to color less

Red color
go out

Acid-alcohol

Red color

mycobacterium

Affinity between carbol fuchsine and mycolic acid
is larger than acid-alcohol and carbol fuchsine

Red color can't
go out
(strong bound
with wax)

5- Wash the slide under tap water (immediately)

(to stop action of acid – alcohol and avoid over decolorizing)

Over decolorization

colorless

Bacillus

Still colorless

Over decolorization

Red color

mycobacterium

Become colorless

colorless

Down
decolorization

Bacillus

Become red

Red color

Down
decolorization

mycobacterium

Still red

حسب التركيز بتحدد الزمن

6- wash the slide with counter stain (methylene blue) for 1 min

Methylene blue

Color less

Bacillus

Become Blue

Methylene blue

Red Color

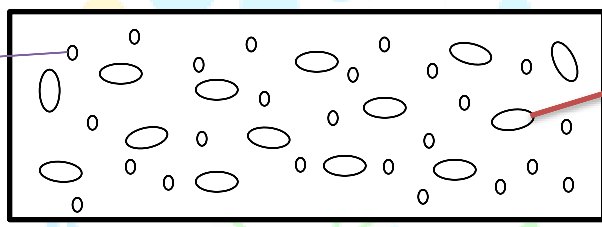
mycobacterium

Still red

7- Wash the slide under tap water (to remove excessive)

8- Blot the slide by filter paper

Mycobacterium
(red)



Bacillus (blue)

بالنتيجة النهائية ما بصير يتبدلو
الالوان بين البكتيريا

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* Example of error *

* **Color less** → Mistake in cooling
  mycobacterium

* **violet** → Down decolorizing (mix between carbol and methylene)
  Bacillus

* **blue** → Over decolorizing
  mycobacterium

Blue/red → Old culture بس تكون الوانهم نص نص – هون الخطأ

Structural stain

* Endospore stain

Binary fusion is the reproductive way in bacteria

Bacteria

- Spore former
  - endospore → Still until cell death
  - exospore → Exit to environment while bacteria still a live
- Non - Spore former

*spore : dormant, non-reproductive

*form spore in stationary phase (in bad condition)

*spore : one genetic matinal with coat (single DNA)

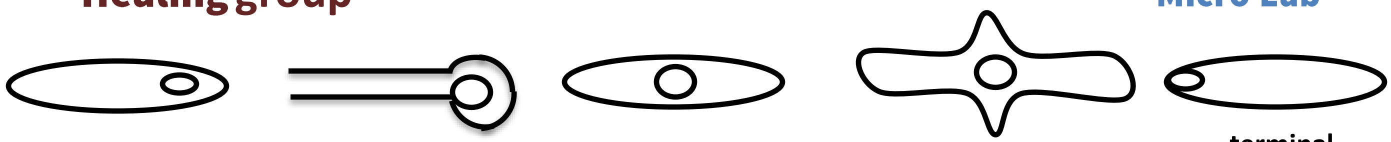
---

* we classified spore to 3 categories :

1- surface of coat
- smooth
- rough

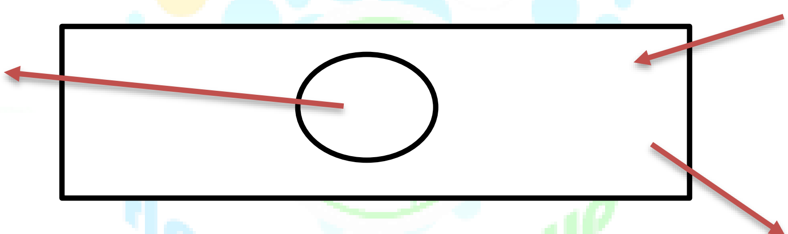
2-shape (position of spore inside the cell)
- Terminal
  - swell
  - Without swell
- central
- subterminal

3- color



1- smear preparation up to air drying (Bacillus) in stationary phase more than 24 hours .

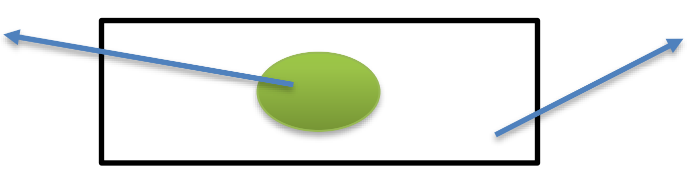
2- Load the slide with <u>primary</u> stain (malachite green) under steaming for <u>not</u> more 7 sec



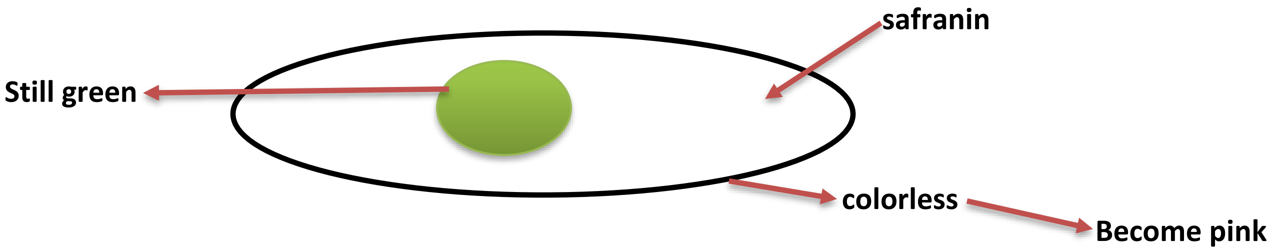
*To enter primary stain to the spore you must use heat
(which come to the coat and try to open it)
So heat increase diameter pores of spore

3- Cool slide at <u>room temp</u> until 5 min (to reclose pores) we sealed (حجز) stain

4- Wash the slide under tap water (decolorization solution)



5- Load the slide with counter stain (safranin) for 1 min



6- Wash your slide under tap water to remove excessive

7- Bolt with filter water

Example errors

* <u>spore</u> (color less) → mistake in cooling

* <u>vegetative</u> (color less) → mistake in wash under tap water

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<u>Please forgive us for any mistake!</u>

Healing group (Nursing group)

تم بحمد الله